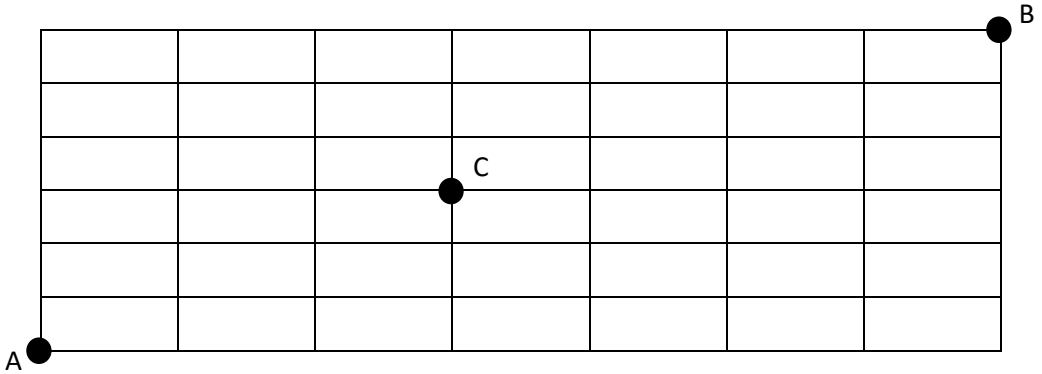
1. Eight different gifts are to be distributed among 12 children. How many distinct results are possible if no child is to receive more than one gift?

2. Delegates from 9 countries, including Russia, France, England, and the United States, are to be seated in a row. How many different seating arrangements are possible if the French and English delegates are to be seated next to each other and the Russian and U.S. delegates are not to be next to each other?

3. Consider the 6x7 grid shown below. Suppose that, starting from the point labeled A, you can go one step up or one step to the right at each move. This procedure is continued until the point labeled B is reached.

- **a.** How many different paths from A to B are possible? *(Hint: Note that to reach B from A, you must take 7 steps to the right and 6 steps upward.)*
- **b.** How many different paths are there from A to B that go through point C?



4. If there are 12 strangers in a room, what is the probability that no two of them celebrate their birthday in the same month?

5. A closet contains 12 pairs of shoes. If 6 shoes are randomly selected, what is the probability that there will be

- **a.** no complete pair?
- **b.** exactly one complete pair?

6. A crime is committed by one of two suspects, A and B. Initially, there is equal evidence against both of them. In further investigation at the crime scene, it is found that the guilty party had a blood type found in 10% of the population. Suspect A does match this blood type, whereas the blood type of suspect B is unknown.

- **a.** Given this new information, what are the odds that A is the guilty party?
- **b.** Given this new information, what are the odds that B's blood type matches that found at the crime scene?

7. Two coins are in a hat. The coins look alike' but one coin is fair (with probability ½ of Heads), while the other coin is biased, with probability ¼ of Heads. One of the coins is randomly pulled from the hat, without knowing which of the two it is. Call the chosen coin "Coin C".

- **a.** Coin C is tossed twice, showing Heads both times. Given this information, what is the probability that Coin C is the fair coin?
- **b.** Are the events "first toss of Coin C is Heads" and "second toss of Coin C is Heads" independent? Explain.
- **c.** Find the probability that in 10 flips of Coin C, there will be exactly 3 Heads. (The coin is equally likely to be either of the 2 coins; do not assume it already landed Heads twice as in (a).)

8. Players Alice and Brian take turns in answering trivia questions, starting with Alice answering the first question. Each time Alice answers a question, she has probability 0.6 of getting it right. Each time Brian plays, he has probability 0.7 of getting it right.

- **a.** If Alice answers 5 questions, what is the probability mass function of the number of questions she gets right?
- **b.** If Alice answers 5 times and Brian answers 4 times, what is the probability mass function of the total number of questions they get right (you can leave your answers as a sum)?
- **c.** Suppose that the first player to answer correctly wins the game (with no predetermined maximum number of questions that can be asked). Find the probability that Alice The odds of an event A are defined by wins the game.

9. A forest contains 25 elk, of which 6 are captured, tagged, and then released. A certain time later, 5 of the 25 elk are captured. What is the probability that at least 2 of these 5 have been tagged? <u>What assumptions are you making?</u>

10. An ectopic pregnancy is twice as likely to develop when the pregnant woman is a smoker as it is when she is a nonsmoker. If 32 percent of women of childbearing age are smokers, what percentage of women having ectopic pregnancies are smokers?

11. Is it possible that an event is independent of itself? If so, when?

12. In how many ways can 8 people be seated <u>at a round table</u> if

- **a.** Ally and Bob must sit next to each other?
- **b.** Chrys and Daniel must not sit next to each other?
- **c.** parts (a) and (b) should hold at the same time? (That is, Ally and Bob must sit together AND Chrys and Daniel must sit apart from each other.)

13. A pair of dice is rolled. What is the probability that the second die lands on a higher value than does the first?

14. Find the probability that the total after rolling 4 fair die is 21.

15. Consider an urn containing 12 balls, of which 8 are white. A sample of size 4 is to be drawn

a. with replacement.
b. without replacement.

What is the conditional probability in each case that the first and third balls drawn will be white given that the sample drawn contains exactly 3 white balls?

16. A city with 6 districts has 6 robberies in particular week. Assume the robberies are located randomly, with all possibilities for which robbery occurred where equally likely. What is the probability that some district had more than 1 robbery?

17. A laboratory blood test is 0.98% effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1% of the healthy persons tested. If 0.5% of the population actually has the disease, what is the probability a person has the disease given that his test result is positive?

- **SOLUTIONS –**

18. Eight different gifts are to be distributed among 12 children. How many distinct results are possible if no child is to receive more than one gift?

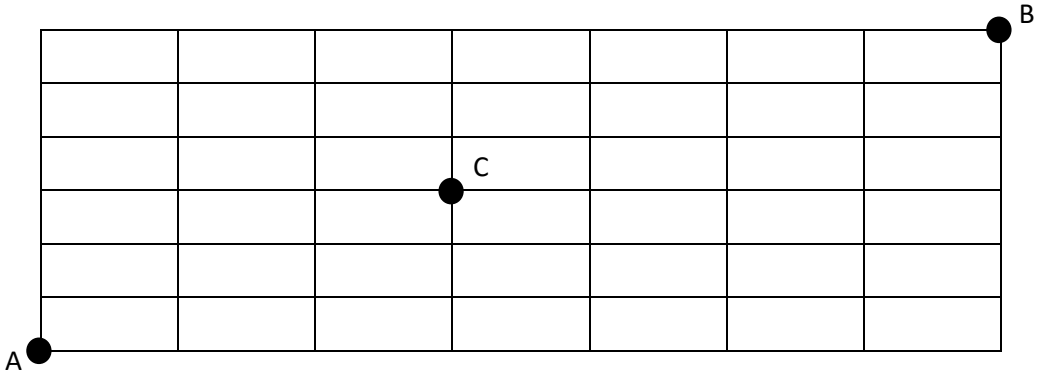
$$\binom{12}{8} 8!$$

19. Delegates from 9 countries, including Russia, France, England, and the United States, are to be seated in a row. How many different seating arrangements are possible if the French and English delegates are to be seated next to each other and the Russian and U.S. delegates are not to be next to each other?

$$8! * 2 - 7! * 2 * 2$$

20. Consider the 6x7 grid shown below. Suppose that, starting from the point labeled A, you can go one step up or one step to the right at each move. This procedure is continued until the point labeled B is reached.

a. How many different paths from A to B are possible? *(Hint: Note that to reach B from A, you must take 7 steps to the right and 6 steps upward.)*
b. How many different paths are there from A to B that go through point C?



a) $\binom{13}{6}$

b) $\binom{6}{3}\binom{7}{3}$

21. If there are 12 strangers in a room, what is the probability that no two of them celebrate their birthday in the same month?

$$\frac{12!}{12^{12}}$$

22. A closet contains 12 pairs of shoes. If 6 shoes are randomly selected, what is the probability that there will be

- **a.** no complete pair?
- **b.** exactly one complete pair?

a)

$$\frac{24 * 22 * 20 * 18 * 16 * 14}{\binom{24}{6}}$$

b)

$$\frac{(24 * 1) * \ 22 * 20 * 18 * 16}{\binom{24}{6}}$$

23. A crime is committed by one of two suspects, A and B. Initially, there is equal evidence against both of them. In further investigation at the crime scene, it is found that the guilty party had a blood type found in 10% of the population. Suspect A does match this blood type, whereas the blood type of suspect B is unknown.

- **a.** Given this new information, what are the odds that A is the guilty party?
- **b.** Given this new information, what are the odds that B's blood type matches that found at the crime scene?

a)

$$P(A|M^A) = \frac{P(M^A|A)P(A)}{P(M^A|A)P(A) + P(M^A|B)P(B)} = \frac{1 * \frac{1}{2}}{1 * \frac{1}{2} + \frac{1}{10} * \frac{1}{2}} = \frac{10}{11}$$

$$\frac{P(A|M^A)}{P(A^C|M^A)} = \frac{10/11}{1/11} = 10{:}1 \ in\ favor\ that\ A\ is\ guilty$$

b)

$$P(M^B|M^A) = \ P(M^B|A, M^A)P(A|M^A) + P(M^B|B, M^A)P(B|M^A)$$
$$= \frac{1}{10} * \frac{10}{11} + 1 * \frac{1}{11} = \frac{2}{11}$$

$$\frac{P(M^B|M^A)}{P\left(M^{B^C}\middle|M^A\right)} = \frac{2/11}{9/11} = 2:9$$

$9:2\ against\ that\ B\ mathes$

24. Two coins are in a hat. The coins look alike' but one coin is fair (with probability ½ of Heads), while the other coin is biased, with probability ¼ of Heads. One of the coins is randomly pulled from the hat, without knowing which of the two it is. Call the chosen coin "Coin C".

- **a.** Coin C is tossed twice, showing Heads both times. Given this information, what is the probability that Coin C is the fair coin?
- **b.** Are the events "first toss of Coin C is Heads" and "second toss of Coin C is Heads" independent? Explain.
- **c.** Find the probability that in 10 flips of Coin C, there will be exactly 3 Heads. (The coin is equally likely to be either of the 2 coins; do not assume it already landed Heads twice as in (a).)

a)

$$P(fair|HH) = \frac{P(HH|fair)P(fair)}{P(HH)} = \frac{\left(\frac{1}{4}\right)\left(\frac{1}{2}\right)}{\left(\frac{1}{4}\right)\left(\frac{1}{2}\right) + \left(\frac{1}{16}\right)\left(\frac{1}{2}\right)} = \frac{4}{5}$$

b) They are not independent. If the first toss is H, then it is an evidence in favor of the fair coin.

c)

$$P(3H|fair).P(fair) + P(3H|biased).P(biased) = \binom{10}{3}\left(\frac{1}{2}\right)^{10}\frac{1}{2} + \binom{10}{3}\left(\frac{1}{4}\right)^{3}\left(\frac{3}{4}\right)^{7}\frac{1}{2}$$

25. Players Alice and Brian take turns in answering trivia questions, starting with Alice answering the first question. Each time Alice answers a question, she has probability 0.6 of getting it right. Each time Brian plays, he has probability 0.7 of getting it right.

- **a.** If Alice answers 5 questions, what is the probability mass function of the number of questions she gets right?
- **b.** If Alice answers 5 times and Brian answers 4 times, what is the probability mass function of the total number of questions they get right (you can leave your answers as a sum)?
- **c.** Suppose that the first player to answer correctly wins the game (with no predetermined maximum number of questions that can be asked). Find the probability that Alice The odds of an event A are defined by wins the game.

a)

$$P(X = k) = \binom{5}{k}(0.6)^k(0.4)^{5-k}$$

b)

$$P(X=k)=\sum_{j=0}^{k}\binom{5}{j}(0.6)^{j}(0.4)^{5-j}\binom{4}{k-j}(0.7)^{k-j}(0.3)^{4-k+j}$$

c) $P(Alice\ wins)=r$

$$r=(0.6)*1+(0.4)*(0.7)*0+(0.4)*(0.3)*r$$

$$r=\frac{15}{22}$$

26. A forest contains 25 elk, of which 6 are captured, tagged, and then released. A certain time later, 5 of the 25 elk are captured. What is the probability that at least 2 of these 5 have been tagged? <u>What assumptions are you making?</u>

$$\sum_{k=2}^{5}\frac{\binom{6}{k}\binom{19}{5-k}}{\binom{25}{5}}$$

We assume that each elk has the equal chance of being caught in both captures. That is, we need to wait long enough after the first release so that elks are dispersed through-out the forest before the second capture.

27. An ectopic pregnancy is twice as likely to develop when the pregnant woman is a smoker as it is when she is a nonsmoker. If 32 percent of women of childbearing age are smokers, what percentage of women having ectopic pregnancies are smokers?

$P(S)=0.32$
$P(N)=1-0.32=0.68$

$$\frac{P(E|S)}{P(E|N)}=\frac{2p}{p}$$

$$P(S|E)=\frac{P(E|S)P(S)}{P(E|S)P(S)+P(E|N)P(N)}=\frac{2p(0.32)}{2p(0.32)+p(0.68)}=\frac{0.64}{0.64+0.68}\sim 0.48$$

28. Is it possible that an event is independent of itself? If so, when?

$P(A) = P(A \cap A) = P(A).P(A) = P(A)^2$

$P(A) = P(A)^2$ only for $P(A) = 0\ or\ 1$.

So it is possible for event with probability 0 or 1.

29. In how many ways can 8 people be seated at a round table if

- **a.** Ally and Bob must sit next to each other?
- **b.** Chrys and Daniel must not sit next to each other?
- **c.** parts (a) and (b) should hold at the same time? (That is, Ally and Bob must sit together AND Chrys and Daniel must sit apart from each other.)

a. $\frac{2!7!}{7} = 2!\,6!$

b. $7! - 6!\,2!$

c. $6!\,2! - 5!\,2!\,2$

30. A pair of dice is rolled. What is the probability that the second die lands on a higher value than does the first?

A= { (1,2), (1,3), (1,4), (1,5), (1,6),
(2,3), (2,4), (2,5), (2,6),
(3,4), (3,5), (3,6),
(4,5), (4,6),
(5,6)}

$N(A) = 5 + 4 + 3 + 2 + 1 = 15$

$$P(A) = \frac{N(A)}{N(S)} = \frac{15}{36}$$

31. Find the probability that the total after rolling 4 fair die is 21.

The rolls (without considering their order) with a total of 21 are: {(6,6,6,3), (6,6,5,4), (5,5,5,6)}

Each have 4! ordering with some symmetric solutions.
Total number of different orderings are:

$\frac{4!}{3!} + \frac{4!}{2!} + \frac{4!}{3!} = 4 + 4 + 12 = 20 = N(A)$

$N(S) = 6^4\ therefore\ P(A) = \frac{20}{6^4}$

32. Consider an urn containing 12 balls, of which 8 are white. A sample of size 4 is to be drawn

a. with replacement.

b. without replacement.

What is the conditional probability in each case that the first and third balls drawn will be white given that the sample drawn contains exactly 3 white balls?

E={1st and 3rd balls are white}
F={exactly 3 white balls are drawn}

$P(E|F) = ?$

a.

$$P(F) = \frac{\binom{8}{3}\binom{4}{1}}{\binom{12}{4}}$$

$$P(EF) = \frac{8 * 7 * 6 * 4}{12 * 11 * 10 * 9} * 2$$

$$P(E|F) = \frac{P(EF)}{P(F)} = \frac{1}{2}$$

b.

$$P(F) = \frac{8 * 8 * 8 * 4}{12 * 12 * 12 * 12}$$

$$P(EF) = \frac{8 * 8 * 8 * 4}{12 * 12 * 12 * 12} * 2$$

$$P(E|F) = \frac{P(EF)}{P(F)} = \frac{1}{2}$$

33. A city with 6 districts has 6 robberies in particular week. Assume the robberies are located randomly, with all possibilities for which robbery occurred where equally likely. What is the probability that some district had more than 1 robbery?

It means that at least one district had no robbery. = 1- each had exactly one robbery

$$= 1 - \frac{6!}{6^6}$$

34. A laboratory blood test is 0.98% effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1% of the healthy persons tested. If 0.5% of the population actually has the disease, what is the probability a person has the disease given that his test result is positive?

E = {person has disease}

F={test is positive}

$P(F|E) = 0.98$

$P(F|E') = 0.01$

$P(E) = 0.005$

$P(E|F) = ?$

$$P(E|F) = \frac{P(F|E)P(E)}{P(F|E)P(E) + P(F|E')P(E')} = \frac{(0.98)(0.005)}{(0.98)(0.005) + (0.01)(0.995)}$$